

# Exercise Sheet 10

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## Cascades and Temporal Networks

5942UE Network Science

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# 10.A Cascades and Threshold Rules

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# Exercise 10.A.1: Threshold Cascade by Hand

Graph (edges  $A-B$ ,  $B-C$ ,  $A-D$ ,  $C-E$ ,  $D-F$ ) with thresholds:  $A(0.4)$ ,  $B(0.5)$ ,  $C(0.3)$ ,  $D(0.5)$ ,  $E(0.5)$ ,  $F(0.3)$

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Seed at  $t = 0$ :  $A$ ,  $C$ . Synchronous update.

1. Compute the degree of each non-seed node.
2. Trace which nodes adopt at  $t = 1, 2, 3$ .
3. Does the cascade reach all nodes?

# Exercise 10.A.2: Threshold Heterogeneity

Same graph as 10.A.1. Two experiments:

- **A:** All thresholds raised to 0.6.
- **B:** Only  $B$ 's threshold raised to 0.6 (others as in 10.A.1).

For seed  $A, C$ :

1. Does the cascade reach all nodes in (A)?
2. Does the cascade reach all nodes in (B)?
3. Why does a single threshold change "block" the cascade in (A)?

## Exercise 10.A.3: Cascade Simulation

Simulate threshold cascades on `nx.barabasi_albert_graph(100, 3)`.

1. At what seed size does the cascade reliably sweep most of the network?
2. Raise threshold to 0.5 — how does final adoption change?
3. Compare seeding **hubs** (top- $k$  degree) vs **random**.

# Exercise 10.A.4: Bridges Differ for Simple vs. Complex Contagion

Dumbbell: two 5-cliques 1..5 and 6..10 joined by bridge 5–6. Node 1 adopts first.

1. **Simple contagion** ( $\beta = 1$ ): does it spread to clique B?
2. **Complex contagion** ( $\theta = 0.4$ ): does it cross the bridge?
3. What change would let the cascade cross?

# 10.B Temporal Networks

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# Exercise 10.B.1: Order-Dependent Reachability

Three nodes  $A, B, C$  with two timed edges:

1. Can information travel  $A \rightarrow C$ ? Trace the path.
  2. Can it travel  $C \rightarrow A$ ?
- $A-B$  exists at  $t = 1$  only
  - $B-C$  exists at  $t = 2$  only
3. Reverse the timing ( $A-B$  at  $t = 2$ ,  $B-C$  at  $t = 1$ ). Does  $A \rightarrow C$  still work?
  4. What does this say about temporal vs. static reachability?

# Exercise 10.B.2: Static Aggregation Hides Causal Order

A hospital contact network over 3 days:

| Day | Edge            |
|-----|-----------------|
| 1   | Nurse–Patient A |
| 2   | Nurse–Patient B |
| 3   | Doctor–Nurse    |

1. Can infection spread from Patient A to Doctor? Trace it.
2. Can it spread from Doctor to Patient A?
3. What might static contact tracing wrongly conclude?

# Exercise 10.B.3: Temporal Reachability in Code

Implement BFS over a temporal edge list.

1. From node 0, which nodes are reachable and at what earliest time?
2. From node 3, is node 0 reachable?
3. Add edge  $(3, 0, 5)$ . How does reachability change?

# Exercise 10.B.4: When Timing Kills Spreading

Star with hub  $H$  and leaves  $A, B, C, D$ . Disease starts at  $A$  at  $t = 0$ .

- **Scenario X:** all edges active at  $t = 1$
  - **Scenario Y:**  $H-A$  at  $t = 1$ ,  $H-B$  at  $t = 2$ ,  $H-C$  at  $t = 3$ ,  $H-D$  at  $t = 4$
1. Which nodes get infected in (X)?
  2. In (Y)?
  3. Now suppose the disease has a **1-step infectious window**. In (Y), does it spread beyond  $H$ ?
  4. What does this illustrate about epidemic containment?

# Wrap-Up

- threshold cascades depend on dense local reinforcement, not just connectivity
- bridges are good for simple contagion but bad for complex cascades
- temporal networks make reachability asymmetric and order-dependent
- contact timing relative to infectious windows can decisively contain spread